Chapter 10
Style Production Control

Introduction

Use Style Production Control to:

- Record the real world implementation of the production plan.
- Manage production orders.
- Make transaction bookings.
- Monitor the progress of orders through production by logging the status of each order at a machine. The order status may be one of the following:
    - Suggested.
    - Planned.
    - Confirmed.
    - Released.
    - Active.
    - Completed.
    - Cancelled.
- Record the following events:
    - Issue of materials.
    - Backflushing of bulk issues.
    - Scrap and re-work .
    - Placing unused materials back in stock.
- Manage machines, work-in-progress inventory (WIP), and inventory stocks.
- Generate production related reports giving costing and efficiency information.

Tip: The accuracy of the reports is dependent on maintaining good discipline in booking operations. Try to balance rigorous discipline with the demands of the production environment.

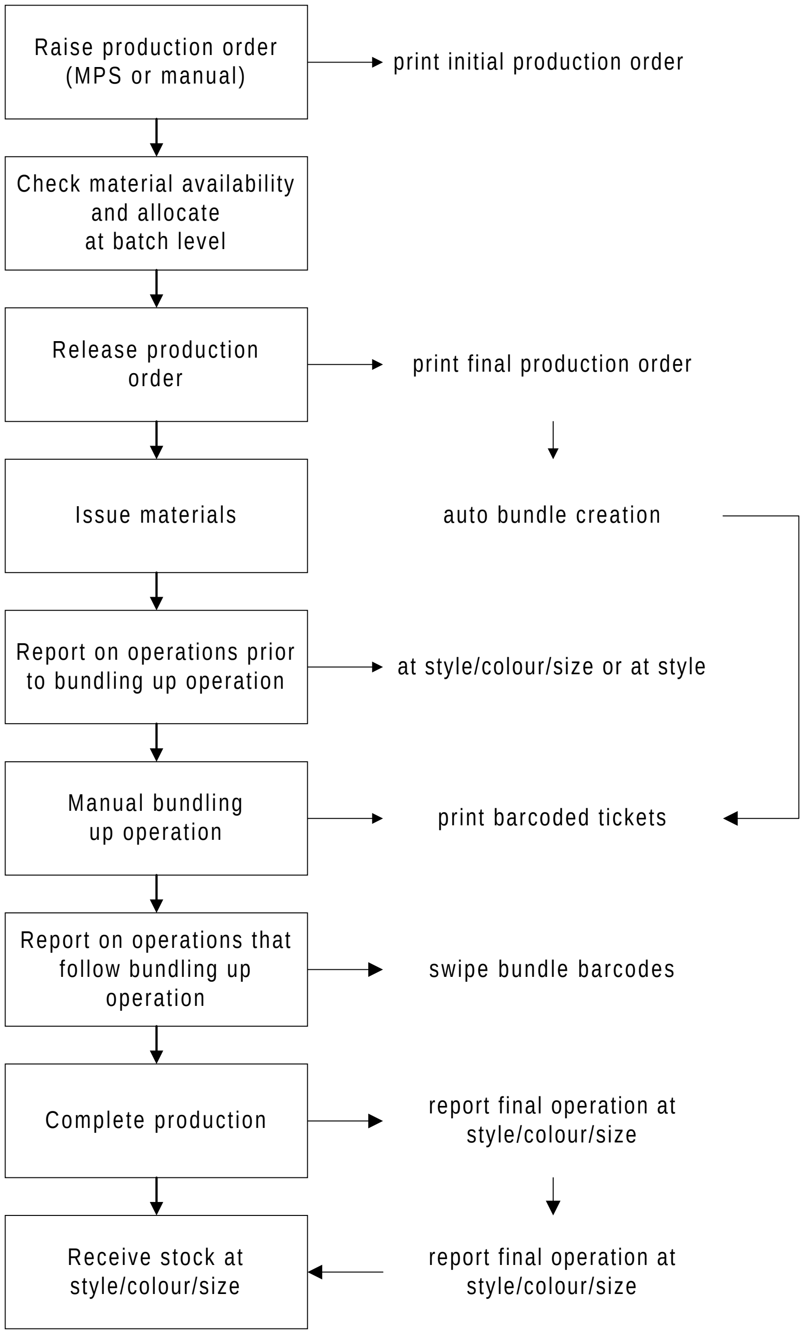

Figure -: Production Control

Production Orders

A production order is a document that details the manufacture of a specific quantity of one style. The order provides authority for carrying out the manufacturing tasks required.

Production orders originate from two sources:

- MPS/MRP suggested orders. MPS/MRP suggested orders are literally orders with quantities and dates suggested by MPS (Master Production Scheduling application) and MRP (Material Requirements Planning application). Such orders are confirmed within those applications and may be subsequently maintained within Production Control. Suggested orders as such are not visible within Production Control.
- Manually created orders. Manually created orders are orders raised within Production Control. An order may be either of the following types:
  - Standard. Based on a standard pre-defined style route. All MPS/MRP suggested orders are of this type. You can also enter them manually.
  - Amended Standard. Where modifications to a standard route have been made to satisfy a particular demand or circumstance.

The life cycle of an order passes through the stages shown below. Each status is identified on the system either by an alpha code or a numeric code. Relevant codes are shown in square brackets in the subsequent expanded definitions.

- Suggested.
- Planned.
- Confirmed.
- Released.
- Active.
- Complete.
- Cancelled.

Suggested [SUG]

This is an order suggested by MPS or MRP processing. A confirmation option changes the status to 'Confirmed' A status of 'Suggested' cannot be entered manually. These orders only appear in the Style Master Production Scheduling and Style Material Requirements Planning applications. The orders are for a specific style and stockroom and have a planning route code, due date and a quantity.

Planned [PLN, 1]

At this point, the order has material and resource requirements based on the selected route/BOM. A planned order represents an intended supply of variants of a finished style. However, at this stage, the order does not create any demand for materials. The orders are for specific styles and stockrooms, and have a planning route code, due date and a quantity. Such orders are shown as 'On

Order' in the Style Inventory Management application. A planned order must subsequently be confirmed.

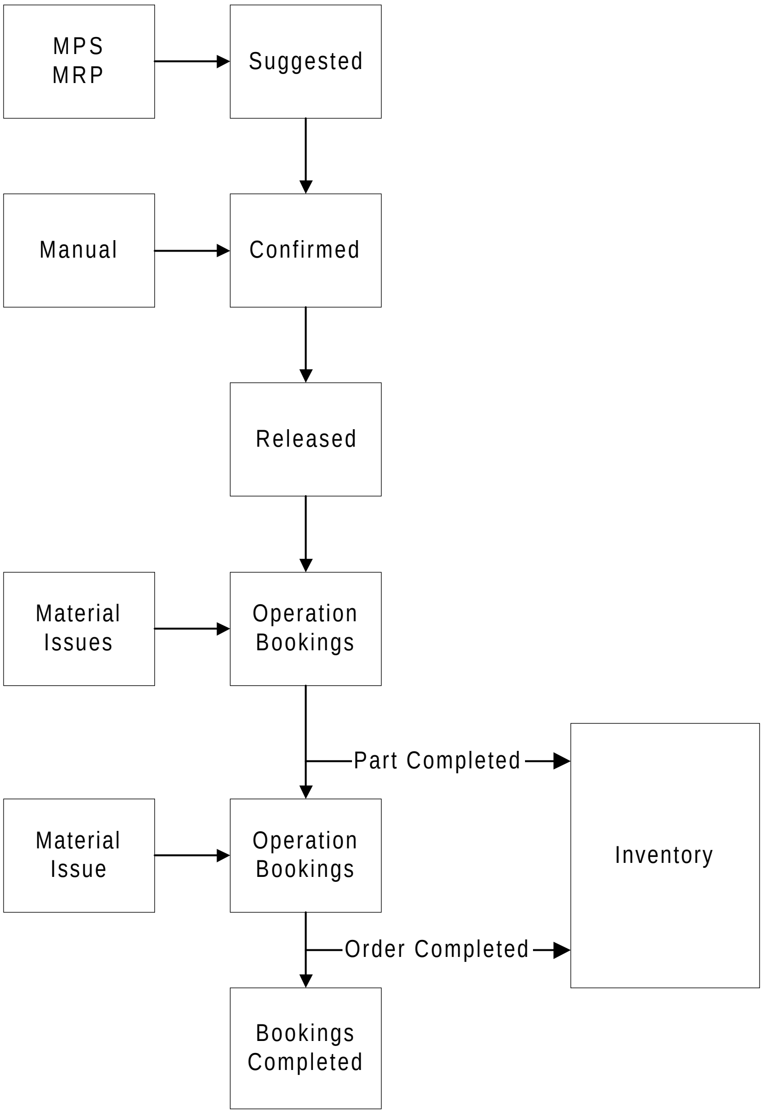

Figure -: Production Order Life Cycle

Confirmed [CON, 2]

Order confirmation causes the allocation of materials in the planned issuing stockroom and/or warehouse if the Allocation of Materials At flag on the Company Profile is set to 1 (confirmation). For standard orders, the system assumes details as defined on the specified route. For amended standard orders, the system uses the standard definition by default, but specific requirements where the standard has been overridden.

The orders have the same characteristics as 'Suggested' orders but with full route/BOM details and resource requirements.

Released [REL, 3]

Order release causes the allocation of materials in the planned issuing stockroom and/or warehouse if the Allocation of Materials At flag on the Company Profile is set to 2 (release) Release prohibits quantity changes although dates and item detail changes may be made. Issues, order and operator bookings, timesheet and bundle booking and finished item receipts may now be performed.

Active [ACT, 4]

The order becomes active as soon as any production activity is recorded against it.

Complete [FIN, 8]

An order may be completed after it is released without necessarily reporting any activity. This procedure may be reversed to return the order to ‘Active’ status, or ‘Released’ if no bookings were reported. Optionally, an order may be set to completed when the planned quantity is reported, although it is possible to continue to receive more than that planned. Alternatively, it is possible to complete an order without receiving the full planned quantity.

Once an order has reached ‘Complete’ status it may be archived and purged from the system via the Select Orders to Archive utility. No material issues or bookings are allowed to these orders.

Cancelled [CNL, 9]

An order may be cancelled prior to the allocating of materials (that is, at Confirmation or Release) by deleting it. It is then available for purging and archiving. These orders are not accessible and therefore cannot be maintained in any way.

Maintaining Orders

Production orders originating in Master Production Scheduling and/or Material Requirements Planning may be maintained; together with any orders created specifically within Production Control.

Maintenance facilities are also available for individual orders within:

- Production release option.
- MPS Review (Master Production Scheduling).
- MRP Review (Material Requirements Planning).

Maintenance may require the adjustment of due or start dates, or quantities, or changes to the associated route/BOM Certain levels of amendment are validated against the order status.

Any maintenance activity which changes the status of the order will generate a transaction.

Automatic Calculation of Start Date

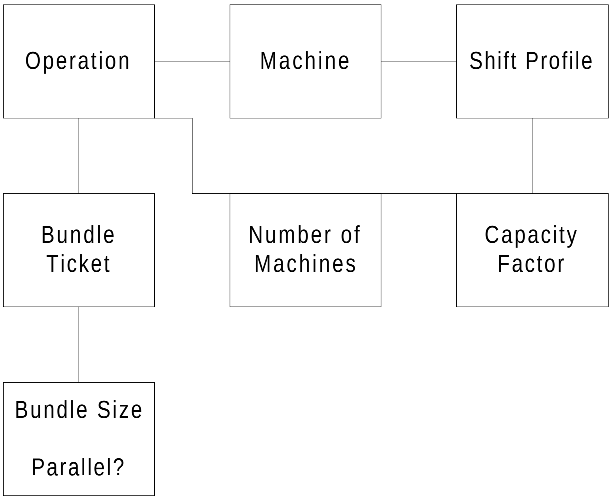

Figure -: Production Order Start Date Calculation

Style Production Control can calculate the start date of an order automatically, as shown below:

Calculate Available Shift Hours Per Day

Shift profile hours

OR

Total of shift lengths

OR

Default to 8 hours

The available hours are determined by the Machine definition. (See Production Definition Management product guide)

Calculate Number Of Machines To Be Used In Parallel

Lowest of:

Number of machines on operations

OR

Order quantity/Default bundle quantity

OR

Capacity factor if shift profiles exist

OR

Shift capacity/Shift length

Available Hours On Machine For The Order =

Available shift hours * Number of machines

Completing Orders

When all work on a production order at operation and/or order level has been completed, then the order can be set to a status of Completed. At such a status, you can archive the order's details.

Individual operations may be flagged as completed as part of booking options or via the Order Completion facility; such operations may however be re-opened.

Orders again can be flagged as completed as part of the booking options or via the Order Completion facility. Re-opening an operation will re-open the order.

As part of housekeeping options, receipts of WIP which have been recorded against 'unexpected' purchase order numbers and which have no GRN number may be reported. In addition, closed shipper (despatch) note details may be purged from the system.

Order Documentation

Documentation associated with a production order comprises:

- Production order.
- Bundle tickets.

Production order documentation

The content of the documentation is affected by several Company Profile flag settings:

- The Print Operations on Production Order flag determines whether or not each operation is printed.
- The Print Operation Text on Production Order flag determines whether or not any additional text held against operations is printed.
- The Print Pick Lists flag indicates whether or not materials will be printed.
- The Suppress Backflush Items on Pick List & Production Order flag indicates whether or not bulk issue (backflush) items should be printed on production orders.
- The Suppress Warehouse Items on Pick Lists & Production Order indicates whether or not warehouse controlled items are to be listed on production orders. This flag does not affect pick lists produced by the Warehousing application.

Archiving Orders

The archive functionality enables you to clear redundant orders so that maximum utilisation of data storage resources can be made.

Bundles

Bundles enable a more detailed level of tracking work-in-progress. This is necessary because production orders exist for a single style which is broken down into colours and sizes.

Therefore, to simply state that 210 have completed operation 140 is not much use. You need to state which colours and sizes have been completed at the operation.

If bundles have been created than this information is readily available, not just by colour and size, but also bundle number as well.

Bundling up involves gathering a number of pieces of each of the components (for example, fronts, backs and sleeves). Bundling is usually performed at operations such as the cutting table or at a marry up point (ie the point where a series of parallel operations or processes come together).

Bundles can be created as colours only; sizes only; or colours and sizes, allowing bundles to exist for example at the pre-dye stages of production.

The system allows for the printing of bundle tickets which can include bar-coded stubs for each operation. These will be physically attached to the bundle and each stub can then be used to feed back information to progress the work through each operation and also record the operator who performed the work.

You can create bundles either manually or automatically.

Note: If the Allow Bundle Tracking flag on the Company Profile is set to 0 (no), then you cannot create bundles.

Transactions

You can report (that is, book) work carried out against production orders through a number of similar options.

Production order transactions consist of the following options:

- Issue materials.
- Production order booking.
- Operator booking.
- Manual timesheet booking.
- Bundle timesheet booking.
- Subcontractor WIP shipper.
- Progress at subcontractor.
- Receive from subcontractor.
- Transaction enquiry.
- Transaction audit report.

Material Issues

You can:

- Enter planned and unplanned material issues. (A planned issue is one that is derived from the bill of material for the style.)
- Review the current status of orders.
- Identify and issue, if required, substitute materials.
- Issue materials to subcontractors.

Production Bookings

Use production bookings to book times and quantities against one or more operations on a single order.

A production order booking lets you book any event which occurs in the production area. It is designed to be most efficient where both of the following conditions are met:

- All bookings relating to a machine or department are posted consecutively.
- The prime reference to the bookings is the production order number.

Whilst most activities are recorded against a production order, this is not mandatory. Style supports the recording of activities such as cleaning, maintenance and supervision etc. for subsequent activity and performance analysis.

Operator Bookings

An operator booking allows you to book the same type of information as a production order booking. The only difference is that it is most efficient to use when all bookings for a production order are presented together, and where the activities of individual operators' activities are recorded.

Separate bookings against the same or a range of activity types, operators (or teams) and orders/operations may be actioned whilst retaining the same header details.

Manual Timesheet Booking

A manual timesheet booking allows you to book the same information that can be booked using Production Order Booking.

However, the option is designed to accept data grouped from individual timesheets. You can input individual operators' timesheets where a single elapsed time is entered for a series of activities. For subsequent performance analysis, the actual time is spread pro-rata across all of the activities carried out during the timeframe covered by the timesheet.

Separate bookings against the same or a range of activity types and orders/operations may be actioned. Header details however cannot be maintained within a session.

Bundle Timesheet Booking

Bundle timesheet booking Is similar to Manual Timesheet Booking. However, you can use bundle timesheet booking when the operator processes bundles rather than booking against a production order number.

Separate bookings against the same or a range of activity types may be actioned.

Subcontract Bookings

Style provides the following subcontract booking options:

- Subcontractor WIP shipper: Enables you to record the sending of WIP inventory to a subcontractor.
- Progress at subcontractor: Is used to record the progress from operation to operation of WIP inventory at a subcontractor. It is used where multiple consecutive operations on the same production order are subcontracted.
- Receive from subcontractor: Is used to record the receipt of WIP inventory back from a subcontractor.

Transaction Manager

When you enter a production booking or WIP inventory transaction, the database is not directly updated. Instead, the transaction is placed on the database to await processing by the Transaction Manager. When active, the Transaction Manager regularly reviews the transaction file for any transaction awaiting processing; it then carries out all of the necessary database updating.

The Transaction Manager runs in a subsystem where it exercises a continuing cycle, only becoming dormant after processing all outstanding transactions. It then becomes active once more after 60 seconds and searches for any additional transactions. As the job runs in a subsystem by itself, when activated, it is processed immediately by the system.

Tip: As part of the daily procedures, you should start the Transaction Manager subsystem. Then, initialise the Transaction Manager itself. At the end of the day, stop the Transaction Manager to enable any day-end jobs to be processed.

Note: If the Transaction Manager is not active, any transactions you enter do not get processed until you re-start the Transaction Manager.

Transaction Reviews

Style provides the following transaction review options:

- Transaction enquiry: Provides an on-line capability to review the contents of transactions which have been input to the system.
- Transaction audit report: Provides the ability to produce a report on the transactions which have been input to the system selecting from a range of parameters.

Activity Codes

All bookings are made against an activity type code that has an associated reporting type that identifies the updates effected by the transaction. In addition to reporting WIP available (good), held and scrap quantities at count point operations, the options are used to book a range of other activities:

- Down time.
- Set up time and materials.
- Indirect overheads.
- Rework.
- Off standard hours.
- Labour and machine time at non count point operations.
- Scrap at non count point operations.

Down Time

To book machine and operator (team) down time, you must enter time against an activity type with a reporting type of 05 (down time). Time and costs associated with down time are reviewed on the Down Time Report

Set Up Time And Materials

To book set up labour time, you must enter time against an activity type with a reporting type of 03 (set up). Any materials consumed specifically by the set up process may be entered by adding an unplanned material issue via the Materials window when booking set up time. If materials are not recorded at this point, they do not contribute to the set up cost. These times and costs are reviewed on the Set Up Cost Report.

Indirect Overheads

Indirect overheads are costs incurred within the production environment that are not directly attributable to production. For example, cleaning. As such, they do not contribute to the cost of items; but if such costs become significant they may need to be covered by the direct overheads defined at cost centre level. These overheads are reviewed on the Indirect Overhead Report.

To book indirect overheads against labour, machines and materials, you must enter time against an activity type with a reporting type of 06 (indirect overhead). Any indirect material costs incurred may be entered by adding an unplanned material issue via the Materials window during indirect overhead time booking.

Rework

Two sorts of rework are catered for:

- Rework held WIP. This is where a quantity has been booked as held, possibly as a result of non conformance to quality standards, but has been reworked to comply, and then made ‘good’ or available. To record labour and machine time to complete the work, time must be booked against an activity type with a reporting type of 04 (rework held WIP).

- Rework at non count point operation. Although no quantities may be recorded against non count point operations, obviously, time may still be spent reworking inventory that would otherwise be missed by the backflushing process. To record such labour and machine time, time must be booked against an activity type with a reporting type of 11 (rework at non count point operations).

The Scrap and Rework Analysis Report considers both types as grouped rework. However, if you designate different activity type(s) for each reporting type, then they are reviewed independently via the Rework Activity Code Analysis Report

Off Standard Hours

Work booked against production may be referred to as On standard hours. However, in certain environments, especially those that utilise piece work, time given over to training or retraining in an unfamiliar job, is referred to as Off standard hours and needs to be recorded independently of On standard hours.

The Manual Timesheet Booking option identifies such time when reporting operator timesheet performance as any time other than that booked against an activity code with a reporting type of 01 (production reporting) or 05 (down time). Such activities would usually be entered against an activity type with a reporting type of 07 (rework).

Labour And Machine Time At Non Count Point Operations

Although actual material costs at operations may be recorded directly by referencing material issues functions, actual labour and machine time at non count point operations must be entered specifically against the operation.

To record this time, time must be entered against an activity type with a reporting type of 07 (labour, machine at non count point operation) at the non count point operation. System validation checks ensure that no WIP inventory quantities are entered.

Scrap At Non Count Point Operations

To record the scrapping of WIP inventory at non count point operations, you must book the scrap quantity against an activity type with a reporting type of 34 (scrap at non count point operation). This scrap value then contributes to the actual production costs of the order.

Note: Booking any activity against an order for the first time changes the status of the order to 'Active'

WIP Inventory Management

WIP inventory management comprises a series of booking transactions that enable the adjustment of inventory balances.

Inventory balances at each WIP location are maintained as a consequence of recording production bookings. Inventory levels can also change as a result of other events (eg scrapping inventory; carrying out physical inventory checks).

At times, you may need to modify WIP inventory transactions generated by booking options eg due to changes in circumstances or events affecting inventory after it has been booked.

Inventory booked good at an operation may be found to be faulty subsequently, eg inventory may be left under bright lights, causing colours to fade. In this case, use the Hold Inventory option to prevent movement of the inventory to the next operation until the inventory is reworked, released or scrapped. If you find that inventory is so badly affected or damaged that it has to be scrapped, use Scrap Available Inventory.

To release WIP previously held through WIP Inventory Control options or placed directly into held status at booking time, usually as a QA stage, use Release Inventory.

Inventory may be booked to WIP locations other than standard (ie the WIP location expected on the route), due to it being worked on by different machines with differing WIP locations. In this case, it may be necessary to re-locate inventory back to the standard location in order to effect proper costing and generate movement records. To do this, use Transfer Location.

Further options allow you to reallocate WIP from one production order to another and to adjust the inventory balance at a WIP location.

Note: You cannot manipulate available WIP after the last operation on a route. If you book WIP inventory as held, you have to release the inventory at the last operation.

Note: WIP Inventory Management prompts with a matrix to determine the styles/variants comprising the total quantity.

<u>Calculation of Actual Time</u>

When all productive work against the hours entered has been defined, the system calculates the assumed actual time for each activity. This time is posted on each individual transaction record and is subsequently used in performance analysis. The actual time of is calculated as follows:

$$\frac{\text{Standard Time for Activity}}{\text{Total Standard Time}} \times \left(\text{Elapsed Time - }\left(\text{Off Std Hrs + Down Time}\right)\right)$$

Where:

Standard Time for Activity - Is the standard time for the activity calculated from the production route.

Total Standard Time - Is the sum of all the standard times for the activities recorded on the timesheet.

Elapsed Time - Is the number of hours entered on the timesheet header.

Off Std Hrs - Is the number of labour hours on the timesheet booked to activities other than those with reporting types of 01 (standard production) or 05 (down time)

Down Time - Any time booked against an activity type with a reporting type of 05 (down time)

Resource Load Management

Resource load management comprises the following functions:

- Review Machine Load.
- Manage Machine Load.
- Review Labour Load.
- Maintain Machine Status.
- Daily Factory Loading.

Style Production considers two kinds of resource:

- Machines which may be grouped for analysis into work centres.
- Labour which comprises operators grouped optionally into teams, and which may further be analysed to operator/skill level.

Facilities within Resource Load Management enable you to review the current loading on machines and labour and to reschedule where any capacity overload can be offloaded to underloaded resources.

Reviewing The Performance Of Your Production Facility

In order to review the performance of your production facility, a range of standard reports is available, grouped into three categories:

- Machines.
- Labour.
- Scrap and Rework.

Machines

Although grouped under an umbrella of machine performance reporting, these reports also have labour elements, as the activities covered can often be booked to labour as well as machines:

- Machine Efficiency.
- Machine Utilization.
- Set Up Cost Report.
- Direct Overhead Variance.
- Down Time Report.
- Indirect Overhead Report.

It must also be remembered that the machine definition is open to interpretation, such that a machine need not necessarily represent a mechanical device requiring operators; but may itself be used to represent one or more operators.

Labour

Use the following report to analyse performance with regard to actual and standard hours booked:

- Direct Labour Variance.

Scrap And Rework

These reports enable you to review why WIP inventory has been scrapped and held inventory reworked:

- Scrap and Rework Analysis.
- Scrap Reason Code Analysis.
- Rework Activity Code Analysis.